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## NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

### First Semester M.Tech (CBCS) Degree Examinations, December 2025

#### 25MCS126 / 25CYS121 – ETHICAL HACKING AND NETWORK DEFENSE

(For MCS, CYS)

Duration: 3 Hours]

[Max. Marks: 100

**Note:** Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II** each and **One full** question from **Module – III**.

#### Module – I

|                                                                                                           | Marks | BT* | CO* | PO* |
|-----------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) What is ethical hacking? Explain the role of ethical hackers in the cyber security.                 | 5     | 2   | 1   | 1   |
| b) Explain the types of offline password attacks.                                                         | 5     | 2   | 2   | 1   |
| c) Explain the countermeasures that must be implemented to prevent the cracking of passwords by a hacker. | 10    | 2   | 2   | 1   |
| 2. a) Explain the types of penetration testing.                                                           | 10    | 2   | 1   | 1   |
| b) What is a threat model? Explain the key steps in threat modeling process.                              | 5     | 2   | 1   | 1   |
| c) Explain the ping sweep techniques used in scanning phase.                                              | 5     | 2   | 2   | 1   |
| 3. a) Explain the role of red team, blue team and purple team in the cyber security.                      | 5     | 2   | 1   | 1   |
| b) What is enumeration? Explain the SNMP enumeration technique.                                           | 10    | 2   | 2   | 1   |
| c) Explain the countermeasures that can be taken against port scanning technique.                         | 5     | 2   | 2   | 1   |

#### Module - II

|                                                                                                                               |    |   |   |   |
|-------------------------------------------------------------------------------------------------------------------------------|----|---|---|---|
| 4. a) Explain different types of attacks that can occur at each layer of OSI model.                                           | 10 | 1 | 3 | 1 |
| b) Explain the DHCP starvation attack with a neat diagram.                                                                    | 5  | 2 | 4 | 1 |
| c) Explain session hijacking.                                                                                                 | 5  | 2 | 4 | 1 |
| 5. a) Outline the Denial of Service (DoS) attack. Explain the countermeasures that can be carried out against the DoS attack. | 10 | 2 | 4 | 1 |
| b) Explain the DNS spoofing and its types.                                                                                    | 5  | 2 | 3 | 1 |
| c) Explain the working of Address Resolution Protocol (ARP) and ARP poisoning sniffing technique.                             | 5  | 2 | 3 | 1 |
| 6. a) Explain the functionality of different layers in OSI reference model.                                                   | 10 | 2 | 3 | 1 |
| b) Explain the most common web application threats and their countermeasures.                                                 | 10 | 2 | 4 | 1 |

#### Module - III

|                                                                                          |    |   |   |   |
|------------------------------------------------------------------------------------------|----|---|---|---|
| 7. a) Explain the application-aware firewalls and proxy firewalls.                       | 10 | 2 | 5 | 1 |
| b) What is snort? What are its basic usages? Explain different modes and rules in snort. | 10 | 2 | 5 | 1 |
| 8. a) Compare Stateless vs. Stateful Packet Filtering Firewalls.                         | 10 | 3 | 5 | 1 |
| b) What is NAT? Why is it required? Explain its working.                                 | 10 | 2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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